When Contrastive Decoding Teaches Time: Annotation-Free RL for Reasoning over Temporal Dynamics

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Abstract

Multimodal Large Language Models (MLLMs) excel in static tasks but frequently struggle to capture video temporal dynamics, leading to failures such as causal inconsistencies. Enhancing this capability is severely constrained by the scarcity and high cost of high-quality temporal annotations, which are essential for prevailing Supervised Fine-Tuning and Reinforcement Learning (RL) paradigms. To bypass this bottleneck, we propose CTRL-V (**C**ontrastive **T**emporal **R**einforcement **L**earning for **V**ideo), an annotation-free framework that integrates Contrastive Decoding into a RL loop. By leveraging the inherent causal disparity between forward and reversed video streams, CTRL-V effectively distills a high-fidelity pseudo ground truth to serve as the supervision signal. This derived supervision drives the RL optimization process via intrinsic temporal alignment, establishing a self-reinforcing cycle that iteratively refines temporal reasoning. Extensive experiments across four temporal reasoning benchmarks show that our CTRL-V consistently improves temporal robustness, achieving superior performance in a fully annotation-free manner. Crucially, systematic analysis confirms that CTRL-V elicits latent temporal potential by aligning intrinsic knowledge with causal dynamics. Our code is available at this link.

Figure 1. To tackle the labeling issue, we transform the model's inference-time discriminability into a permanent parameter optimization. As observed (Bottom), contrastive decoding effectively disentangles temporal ambiguity, serving as a high-quality self-supervision signal for solidifying the model's intrinsic capabilities.

1. Introduction

The recent surge in Multimodal Large Language Models (MLLMs) (Caffagni et al., 2024; Wu et al., 2024; Du et al., 2024; Ye et al., 2023; Jiao et al., 2024; Tang et al., 2025) has revolutionized the landscape of vision-language understanding. By bridging visual encoders (Zhao et al., 2024) with powerful Large Language Models (LLMs) (Zhang et al., 2023; Jiang et al., 2024; Touvron et al., 2023; Achiam et al., 2023; Zhang et al., 2022), these systems have demonstrated exceptional proficiency in static visual tasks, such as image captioning (Sidorov et al., 2020; Narins et al., 2023; Chen et al., 2024; Cheng et al., 2025), visual question answering (Yue et al., 2024; Dong et al., 2024; Lu et al., 2023) and visual grounding (Li et al., 2022; Peng et al., 2023; Liu et al., 2024b). However, the transition from static images to dynamic videos introduces a critical dimension: time. Unlike images, robust video understanding requires deciphering complex temporal dynamics and causal dependencies. In this domain, MLLMs often struggle to capture temporal directionality, leading to failures such as temporal hallucinations (Choong et al., 2024; Sun et al., 2024; Li et al., 2025) and causal inconsistencies (Wu et al., 2023; McCrae et al., 2022; Yan et al., 2025). These errors primarily stem from the models' inability to ground their generation in dynamic visual cues. Lacking a robust grasp of sequential causality,

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models fail to distinguish between the actual chronological flow and causally inverse sequences, resulting in severe temporal inconsistencies.

To mitigate such failures, prevailing methodologies primarily resort to Supervised Fine-Tuning (SFT) (Zhang et al., 2024b; Li et al., 2024b; Zhang et al., 2025) or Reinforcement Learning (RL) (Yu et al., 2024; Shao et al., 2024) with high-quality manually annotated data. However, within the domain of video temporal reasoning, both paradigms confront a shared bottleneck: the prohibitive difficulty of acquiring effective external supervision. SFT necessitates the labor-intensive curation of long-form reasoning chains to explicate chronological dependencies. In contrast, although RL approaches increasingly prioritize concise outcomes under the influence of the emerging Reinforcement Learning with Verifiable Reward (RLVR) (Yue et al., 2025; Wen et al., 2025) paradigm, they remain constrained by the high cost of obtaining supervision. In temporal reasoning, validating even a concise answer incurs labor costs comparable to annotating a full trajectory, as both processes require intrinsic comprehension of complex visual dynamics. Consequently, this dependency on external signals constitutes a critical scalability barrier.

To bypass the reliance on external temporal supervision, we exploit the inherent asymmetry between forward and reversed video streams. A temporally grounded model should exhibit directional sensitivity: semantically valid event progressions align with the forward stream, while their reversed counterparts induce causal violations. By contrasting model likelihoods under these two views, we amplify the probabilistic differences between temporally consistent and inconsistent actions, yielding an intrinsic supervision signal that enforces temporal logic without human annotations, as illustrated in Figure 1.

Building on this principle, we introduce CTRL-V (**C**ontrastive **T**emporal **R**einforcement **L**earning for **V**ideo), an annotation-free framework that integrates Contrastive Decoding (CD) into a reinforcement learning loop for intrinsic temporal supervision. CTRL-V contrasts model predictions conditioned on forward and reversed video streams by subtracting their logits, isolating causal temporal cues inherent to the video. As a result, this contrastive distribution produces a pseudo target, $\mathbf{y}_{\text{contrast}}$, which emphasizes physically plausible temporal progression while suppressing causally inconsistent responses. Unlike prior CD approaches (Leng et al., 2024; Lee et al., 2024; Qi et al., 2025; Zhang et al., 2024a; Jung et al., 2025) that operate solely at inference time, this self-generated target is then used to guide reinforcement learning via a model-based temporal evaluation objective that rewards alignment with chronological consistency. As the model improves, the quality of the contrastive targets correspondingly increases, forming a self-reinforcing optimization process that progressively strengthens temporal reasoning without requiring any external annotations.

Empirically, the proposed CTRL-V demonstrates robust efficacy across four temporal reasoning benchmarks, consistently outperforming video MLLMs (Zhang et al., 2024b; Yao et al., 2024; Lin et al., 2024; Liu et al., 2024a; Li et al., 2024b; OpenGVLab, 2024; Li et al., 2024a) and temporal reasoning baselines (Shen et al., 2025; Ren et al., 2024; Cheng et al., 2024; Hua et al., 2025) in fully annotation-free settings. Furthermore, comprehensive ablations and analysis confirm that the proposed contrastive mechanism is pivotal, effectively incentivizing the model's latent potential to align intrinsic knowledge with valid causal dynamics.

Our contributions are summarized as follows:

- We propose **CTRL-V**, an annotation-free RL framework that addresses the supervision bottleneck in video temporal reasoning by exploiting intrinsic contrastive signals. CTRL-V employs a novel CD-driven mechanism to construct temporally rectified pseudo targets, $\mathbf{y}_{\text{contrast}}$, enabling effective self-optimization without external annotations.
- Extensive experiments across multiple video temporal reasoning benchmarks show that our CTRL-V consistently improves temporal robustness and causal consistency, achieving competitive or superior performance in a fully annotation-free setting.
- Through systematic ablations and analysis, we discover that CTRL-V effectively **elicits** latent temporal reasoning capabilities in MLLMs, demonstrating that intrinsic contrastive supervision can align pretrained knowledge with causal video dynamics.

2. Related Work

2.1. Video Temporal Reasoning with MLLMs

Building on static MLLMs (Liu et al., 2023; Alayrac et al., 2022; Li et al., 2023a; Zhu et al., 2024), video MLLMs (Zhang et al., 2024b; Yao et al., 2024; Lin et al., 2024; Liu et al., 2024a; Li et al., 2024b; OpenGVLab, 2024; Li et al., 2024a; Zhang et al., 2025) extend visual reasoning to the temporal dimension, typically enhancing capabilities via data-centric SFT with dense temporal annotations (Shen et al., 2025; Ren et al., 2024; Cheng et al., 2024; Hua et al., 2025). However, this supervision-heavy paradigm faces a critical scalability bottleneck, as curating high-quality temporal instruction data is prohibitively costly (Maaz et al., 2024; Chen et al., 2024). To bypass this constraint, our CTRL-V introduces an annotation-free alternative that exploits the video's intrinsic causal structure via time reversal, enabling self-supervised temporal optimization without incurring external annotation costs.

2.2. Reinforcement Learning

Reinforcement Learning (RL) has evolved significantly (Schulman et al., 2017; Rafailov et al., 2023; Shao et al., 2024), with the recent RLVR (Yue et al., 2025; Wen et al., 2025) paradigm achieving remarkable success in math and code by leveraging cost-free, verifiable outcomes. Nevertheless, extending this to video temporal reasoning confronts a unique supervision bottleneck. Unlike deterministic domains (Guo et al., 2025; Lightman et al., 2023), acquiring supervision for video is inherently costly: verifying even concise temporal conclusions requires comprehensive human assessment of complex dynamics. To overcome this scalability barrier, CTRL-V shifts to intrinsic self-supervision. We exploit the inherent causal asymmetry of time reversal to synthesize a robust reward signal, enabling the model to optimize temporal reasoning without relying on expensive external ground-truth outcomes.

2.3. Contrastive Decoding

Contrastive Decoding (CD) was originally proposed to suppress generic language priors (Li et al., 2023b) and has since been adapted for static vision-language tasks to mitigate object hallucinations (Leng et al., 2024; Lee et al., 2024). In the video domain, recent works (Qi et al., 2025; Zhang et al., 2024a; Jung et al., 2025; Wu et al., 2025; Pang et al., 2025) apply CD solely during inference, contrasting logits between original and heuristically distorted streams (e.g., via chronological shuffling or masking) to filter temporal inconsistencies. However, these inference-time interventions incur prohibitive computational costs by doubling forward passes and serve merely as external correctives rather than addressing intrinsic model deficiencies. Conversely, CTRL-V shifts this paradigm to training-time internalization. By exploiting temporal directionality to distill causal signals for reinforcement learning, we elicit latent temporal reasoning capabilities, yielding a self-contained model that eliminates the need for dual-stream computation at inference.

3. Theoretical Analysis

In this section, we establish the theoretical foundation of the CTRL-V framework. We analyze the inherent limitations of standard video instruction tuning and derive a contrastive objective designed to maximize the discriminability of causal temporal events.

3.1. Preliminary

Standard Contrastive Decoding. Contrastive Decoding (Li et al., 2023b) aims to enhance generation quality by amplifying the output likelihood of an expert policy π_{exp} relative to an amateur policy π_{ama}. The core intuition is to subtract the amateur's errors from the expert's predictions. Formally, the contrastive objective for a token y_t is defined as the difference in log-probabilities:

$$\mathcal{L}_{\text{CD}}(y_t) = \ln \pi_{\text{exp}}(y_t \mid x, y_{<t}) - \ln \pi_{\text{ama}}(y_t \mid x, y_{<t}), \quad (1)$$

To ensure semantic plausibility, it employs an adjustable mask $\mathcal{S}_t$ that retains only tokens within a high-confidence nucleus defined by a hyperparameter $\beta_{CD} \in (0, 1]$:

$$\mathcal{S}_t = \left\{ w \in \mathcal{V} \;\middle|\; \pi_{\text{exp}}(w) \geq \beta_{CD} \cdot \max_k \pi_{\text{exp}}(k) \right\}. \quad (2)$$

This constraint effectively prevents the selection of implausible tokens from the low-probability tail of the expert distribution.

Group Relative Policy Optimization. GRPO is a reinforcement learning algorithm designed to optimize a policy π_θ without requiring an additional value network (critic). For each query q, GRPO samples a group of outputs $\{o_i\}_{i=1}^G$ from the reference policy π_{ref} (or $\pi_{\theta_{\text{old}}}$). The optimization objective maximizes the surrogate gain based on group-normalized advantages:

$$\begin{aligned}\mathcal{J}_{GRPO}(\theta) =& \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(O|q)] \\ & \frac{1}{G}\sum_{i=1}^{G} \frac{1}{|o_i|}\{\min[\frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})}\hat{A}_{i,t}, \\ & \text{clip}(\frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})}, 1-\varepsilon, 1+\varepsilon)\hat{A}_{i,t}] \\ & - \beta\mathbb{D}_{KL}[\pi_\theta || \pi_{ref}]\}.\end{aligned} \quad (3)$$

where $\rho_t^{(i)} = \frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})}$ is the probability ratio, and ε is the clipping parameter. Crucially, the advantage $\hat{A}_t^{(i)}$ is computed locally within the group to reduce variance:

$$\hat{A}_t^{(i)} = \frac{r(o_i) - \mu(\{r(o_j)\}_{j=1}^G)}{\sigma(\{r(o_j)\}_{j=1}^G) + \delta}, \quad (4)$$

where μ and σ represent the mean and standard deviation of rewards within the sampled group.

3.2. Temporal-Causal Contrastive Rectification

Let $\mathcal{D} = \{(V^+, q, y^*)\}$ denote the dataset. We construct a negative $V^- = \Phi(V^+)$ which implies a semantic flow inconsistent with the ground truth y^*. Standard policies often suffer from temporal ambiguity, failing to differentiate V^+ from V^- and assigning high probabilities to y^* even under the inconsistent V^-. To enforce strictly causal discriminability, we define the **Rectified Contrastive Log-Probability** $\mathcal{L}_t(w)$ for any token $w \in \mathcal{S}_t$:

$$\begin{aligned}\mathcal{L}_t(w) =& (1+\alpha)\ln \pi_\theta(w \mid V^+, q, y_{<t}) \\ & - \alpha \ln \pi_\theta(w \mid V^-, q, y_{<t}).\end{aligned} \quad (5)$$

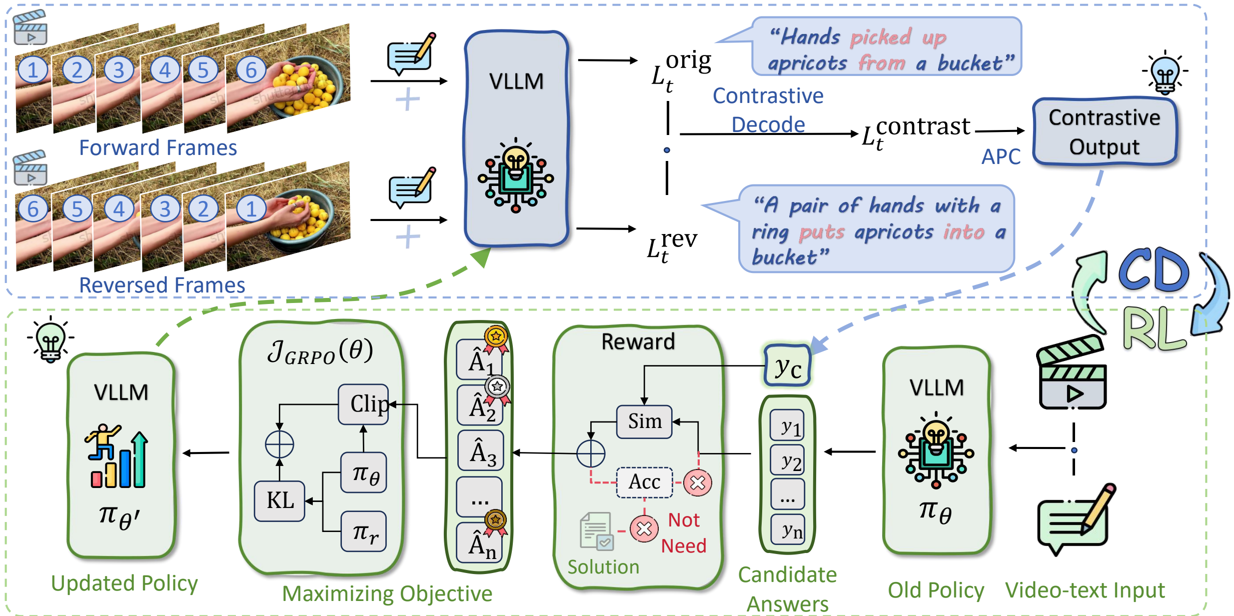

Figure 2. Illustration of the proposed CTRL-V pipeline. We invoke Contrastive Decoding (CD) to exploit the probability shift between forward and reversed video streams, mathematically isolating causal cues to generate a robust pseudo-label $\mathbf{y}_{\text{contrast}}$. This label functions as a self-generated supervision signal for the self-iterative GRPO loop. By optimizing the policy π_θ to align its generated candidates with this intrinsic causal anchor, our CTRL-V effectively elicits latent temporal reasoning capabilities in an annotation-free manner.

where $\alpha > 0$ serves as the penalty coefficient. where $\alpha > 0$ serves as the penalty coefficient. By penalizing the likelihood under V^-, this formulation explicitly suppresses the alternative semantics y^- that are temporally inconsistent with the V^+, and sharpens the model's sensitivity to the temporal directionality.

3.3. Monotonic Expansion of the Decision Margin

We define the **Temporal Decision Margin** $\mathcal{M}(\theta)$ as the expected divergence between the ground truth y^* and the alternative y^-. Substituting Eq. 5 yields the decomposed objective:

$$\mathcal{M}(\theta) \triangleq \mathbb{E}_{V\sim\mathcal{D}}\left[\mathcal{L}_t(y^*) - \mathcal{L}_t(y^-)\right]. \tag{6}$$

$$\mathcal{M}(\theta) = \underbrace{(1+\alpha)\ln\frac{\pi_\theta(y^* \mid V^+)}{\pi_\theta(y^- \mid V^+)}}_{\mathcal{T}_{\text{pos}}} + \underbrace{\alpha\ln\frac{\pi_\theta(y^- \mid V^-)}{\pi_\theta(y^* \mid V^-)}}_{\mathcal{T}_{\text{disc}}}. \tag{7}$$

Here, $\mathcal{T}_{\text{pos}}$ ensures standard prediction accuracy on V^+, while $\mathcal{T}_{\text{disc}}$ explicitly measures the **discriminative capability** to associate the negative context V^- with the alternative hypothesis y^-.

Under GRPO, the policy update θ_{k+1} implicitly maximizes this margin. Assuming monotonic policy improvement, we obtain:

$$\frac{\partial\mathcal{M}(\theta)}{\partial k} \approx (1+\alpha)\frac{\partial\mathcal{T}_{\text{pos}}}{\partial k} + \alpha\frac{\partial\mathcal{T}_{\text{disc}}}{\partial k} > 0. \tag{8}$$

This inequality confirms a self-reinforcing loop: as temporal discrimination ($\mathcal{T}_{\text{disc}}$) improves, the contrastive supervision signal becomes sharper, driving continuous refinement in subsequent iterations.

3.4. Key Insight: Causal Dynamics Excavate Potential

CTRL-V operates on the fundamental premise that the pre-trained model already possesses a *nascent* capability to distinguish temporal directionality. Our framework does not inject new knowledge. Rather, it functions as a resonance chamber that detects and aggressively amplifies this pre-existing discriminability. To rigorously prove this, we analyze the gradient dynamics of the GRPO update. By decomposing the optimization objective, we identify that the update trajectory is strictly modulated by the model's own intrinsic preferences. We derive the **Latent Amplification Gradient Identity**:

$$\begin{aligned}\nabla_\theta\mathcal{J}_{\text{GRPO}} &= \mathbb{E}_{o_i\sim\pi_{\text{old}}}\left[\frac{1}{G}\sum_{i=1}^{G}\nabla_\theta\min\left(\rho_i\hat{A}_i, \text{clip}\,\hat{A}_i\right)\right] \\ &\cong \mathbb{E}_{\tau\sim\pi_\theta}\Bigg[\mathbb{I}_{\text{clip}}(\rho_\tau)\cdot\underbrace{\left(\frac{\pi_\theta(\tau \mid V^+)}{\pi_\theta(\tau \mid V^-)}\right)^{\alpha}}_{\substack{\Psi(\tau;\theta):\\ \text{Intrinsic Excavation Factor}}} \\ &\quad\cdot\nabla_\theta\ln\pi_\theta(\tau \mid V^+)\Bigg].\end{aligned} \tag{9}$$

This derivation reveals a critical dependency: the magnitude of the learning signal $\Psi(\tau;\theta)$ is determined entirely by the model's pre-existing likelihood ratio. For causal events where the model retains a faint trace of inherent understanding (i.e., $\pi(V^+) > \pi(V^-)$), the exponent α non-linearly magnifies this weak signal into a dominant gradient force. In rare cases where the model incorrectly favors the negative context (i.e., the ratio drops below unity), the scalar Ψ automatically attenuates the gradient magnitude. By iteratively solidifying these amplified signals into the parameters, CTRL-V effectively transforms the model's implicit, weak intuition into an explicit, robust cognitive capability, proving that the temporal alignment is excavated from the model's own manifold rather than learned from external supervision.

Algorithm 1 Dynamic Self-Iterative Contrastive RL

Input: Policy π_{θ_0}, video V, query q, iterations K.
Hyperparameters: strength α, group size G.
for $k = 0, 1, \ldots, K-1$ **do**
  $V_{k,\text{rev}} \leftarrow \text{ReverseFrames}(V_k)$
  $\text{logit}_t^{\text{diff}} \leftarrow \text{logit}_{\theta_k}(V_k) - \text{logit}_{\theta_k}(V_{k,\text{rev}})$
  $\mathcal{P}_k(y_t) \leftarrow \text{Softmax}\left(\text{logit}_{\theta_k}(V) + \alpha \cdot \text{logit}_t^{\text{diff}}\right)$
  $\mathbf{y}_k^* \sim \mathcal{P}_k(y \mid q)$
  $\{o_i\}_{i=1}^G \sim \pi_{\theta_k}(y \mid V_k, q_k)$
  $R_i \leftarrow \text{Reward}(o_i, \mathbf{y}_k^*) \in [0,1]$
  $\hat{A}_i \leftarrow \text{Advantage}(\{R_1, \ldots, R_G\})$
  $\theta_{k+1} \leftarrow \theta_k + \eta \nabla_\theta \mathcal{J}_{\text{GRPO}}(\pi_\theta, \hat{A}_i)$
end for
return π_{θ_K}

4. Method

4.1. Video-based Contrastive Decoding

To isolate temporal dynamics, we construct a negative context V_{rev} by strictly reversing the frame order of the original video V, effectively inverting causal logic while preserving spatial features. During the inference phase, the model processes the query q under both the original and reversed contexts in parallel. For each decoding step t, we compute the output logits for both streams as:

$$\begin{aligned} L_t^{\text{orig}} &= \text{logit}_\theta(y_t \mid V, q, y_{<t}), \\ L_t^{\text{rev}} &= \text{logit}_\theta(y_t \mid V_{\text{rev}}, q, y_{<t}). \end{aligned} \tag{10}$$

We then derive a reliable guidance signal $\mathbf{y}_{\text{contrast}}$ by penalizing anti-causal predictions found in the reversed stream. Specifically, we compute the rectified logits $\tilde{L}_t$ using an adaptive contrastive formula:

$$\tilde{L}_{t,w} = \begin{cases} (1+\alpha)L_{t,w}^{\text{orig}} - \alpha L_{t,w}^{\text{rev}} & \text{if } w \in \mathcal{S}_t, \\ -\infty & \text{otherwise}, \end{cases} \tag{11}$$

where $\mathcal{S}_t$ denotes the validity mask ensuring semantic plausibility. The resulting trajectory $\mathbf{y}_{\text{contrast}}$ sampled from $\tilde{L}$ serves as the "excavated" pseudo-ground truth for the current iteration.

4.2. Self-Reinforcement with Dynamic Supervision

To solidify the capability exhibited by $\mathbf{y}_{\text{contrast}}$ into the model parameters, we propose a modified Group Relative Policy Optimization (GRPO) framework tailored for self-supervision, as shown in Figure 2.

First, the model assumes the role of a **Teacher**. Utilizing the contrastive decoding method (Eq. 11), it generates a high-quality pseudo-label $\mathbf{y}_{\text{contrast}}$, which represents the upper bound of the model's current temporal reasoning. Subsequently, the model switches to the **Student** role, sampling a group of G diverse responses $\{o_1, \ldots, o_G\}$ from its standard policy π_θ without contrastive intervention.

To evaluate these candidates, we design a reference-based reward function powered by an external LLM. Since $\mathbf{y}_{\text{contrast}}$ contains the correct causal logic "excavated" from the video, the LLM acts as a judge to score the **temporal consistency** between each student response o_i and the teacher's guidance $\mathbf{y}_{\text{contrast}}$.

This process forms a recursive optimization loop. The policy parameters are updated to maximize the likelihood of highly consistent responses, which is formalized as:

$$\begin{cases} \mathbf{y}_{\text{contrast}}^{(k)} \leftarrow \text{VideoCD}(\pi_{\theta_k}, V, V_{\text{rev}}), & \text{(Supervision)} \\ \theta_{k+1} \leftarrow \text{argmax}_\theta\, \mathcal{J}_{\text{GRPO}}(\theta; \mathbf{y}_{\text{contrast}}^{(k)}). & \text{(Update Policy)} \end{cases} \tag{12}$$

This alternating cycle creates a positive feedback loop: as the policy π_θ improves, the teacher's guidance $\mathbf{y}_{\text{contrast}}$ naturally becomes more precise. By iteratively bootstrapping the model's temporal capability through this evolving supervision, as detailed in Algorithm 1, our framework enable effective self-optimization without requiring manual labels.

5. Experiments

5.1. Benchmarks

We introduce **TemporalQA**, a novel dataset tailored to mitigate the ambiguity inherent in generic video benchmarks (e.g., "Describe the video"). Unlike existing ones that heavily rely on broad inquiries, our TemporalQA employs *sequentially directed* prompts (e.g., "What actions did the subject perform successively?") to rigorously enforce chronological reconstruction without textual answer leakage. Partitioned into training, validation, and test splits, this dataset serves as the primary corpus for our proposed CTRL-V optimization. More details are available in Appendix B.

We evaluate our model on four temporal reasoning benchmarks, including our proposed **TemporalQA** and three

Table 1. Main experimental results on three existing temporal video QA benchmarks, including RTime-QA, TempCompass, MVBench, and our proposed TemporalQA dataset. Our method, CTRL-V, achieves state-of-the-art performance across all datasets in terms of accuracy and overall score metrics. Best results are highlighted in bold.

Method	Base Model	RTime-QA			TempCompass		MVBench		**TemporalQA**	
		Acc	Strict Acc	Score	Acc	Score	Acc	Score	Acc	Score
General Video MLLMs										
LLaVA-NeXT-Video (Zhang et al., 2024b)	Vicuna-7B	69.10	51.34	3.52	57.04	3.22	45.66	2.83	58.26	3.36
MiniCPM-V-2.6 (Yao et al., 2024)	Qwen2-7B	60.83	37.71	3.30	59.73	3.61	47.18	3.00	60.21	3.41
VideoLLaVA (Lin et al., 2024)	Vicuna-7B	58.03	38.20	3.26	45.76	2.66	53.34	2.85	52.75	3.11
LLaVA1.5 (Liu et al., 2024a)	Vicuna-7B	53.89	33.58	2.90	48.55	3.27	38.57	2.72	61.05	3.63
VideoChat2 (Li et al., 2024b)	Mistral-7B	72.38	54.26	3.63	63.62	3.52	55.36	3.36	70.32	3.84
InternVL2 (OpenGVLab, 2024)	InternLM2.5-7B	67.64	48.42	3.61	58.73	3.23	55.40	3.46	63.55	3.66
LLaVA-OneVision (Li et al., 2024a)	Qwen2-7B	55.72	39.42	2.78	63.32	3.49	61.23	3.28	69.15	3.90
Existing Temporal Reasoning Baselines										
LongVU (Shen et al., 2025)	Qwen2-7B	42.94	20.92	2.53	53.34	3.13	64.78	3.63	62.28	3.51
TimeChat (Ren et al., 2024)	LLaMA2-7B	33.70	11.44	2.07	24.9	1.71	27.00	1.45	39.95	2.16
VideoLLaMA2 (Cheng et al., 2024)	LLaMA2-7B	59.73	29.93	3.35	40.67	2.38	49.80	2.59	56.25	2.92
V2Xum-LLM (Hua et al., 2025)	Vicuna-7B	46.35	25.30	2.72	33.83	2.34	48.36	2.86	56.45	3.20
CTRL-V(Ours)	Qwen2.5-7B	**85.16**	**73.48**	**4.29**	**67.56**	**3.72**	**68.03**	**3.95**	**75.76**	**4.04**

Table 2. **Bold** values indicate the best performance within each backbone model group. The results demonstrate that our proposed **CTRL-V** method consistently achieves superior performance gains across various foundational models.

Base Model	Method	RTime-QA			TempCompass		MVBench		**TemporalQA**	
		Acc	Strict Acc	Score	Acc	Score	Acc	Score	Acc	Score
Qwen2.5-VL-7B (Bai et al., 2025b)	Base	68.49	50.36	3.64	62.13	3.61	62.01	3.69	64.87	3.64
	+ CTRL-V	**80.90**	**68.37**	**4.16**	**68.11**	**3.73**	**64.63**	**3.88**	**74.47**	**4.00**
LLaVA-NeXT-Video-7B (Zhang et al., 2024b)	Base	69.10	51.34	3.52	57.04	3.22	45.66	2.83	58.26	3.36
	+ CTRL-V	**76.89**	**62.53**	**3.85**	**64.27**	**3.50**	**49.89**	**3.14**	**68.83**	**3.70**
Qwen3-VL-7B (Bai et al., 2025a)	Base	62.41	41.36	3.62	62.33	3.62	52.53	2.81	66.43	3.76
	+ CTRL-V	**74.57**	**61.56**	**4.05**	**66.02**	**3.79**	**59.65**	**3.59**	**67.66**	**3.78**

established benchmarks: **RTime-QA** (Liu et al., 2025) (adapted to an open-ended format) for atomic event directionality, the caption generation task of **TempCompass** (Liu et al., 2024c) for fine-grained attributes, and **MVBench** (Li et al., 2024b) for capabilities ranging from perception to high-level cognition.

5.2. Implementation Details

We employ Qwen2.5-VL-7B-Instruct as our foundational baseline for main experiments. In contrastive decoding, we set the hyperparameters to $\alpha = 1.0$ and $\beta = 0.2$. We employ GPT-3.5-turbo to assign scores following a standardized rubric to ensure consistent and objective evaluation. On RTime-QA, the reported strict Acc requires a model to correctly answer both questions in a paired sample (i.e., the original video and the reversed video). Only when both answers are correct is the pair counted as a true; otherwise, it is considered as a failure. This metric emphasizes robust temporal reasoning and sensitivity to frame ordering. Further details are provided in the Appendix C.

5.3. Main Results

We compare the proposed CTRL-V against a set of state-of-the-art video MLLMs, including LLaVA-NeXT-Video (Zhang et al., 2024b), VideoChat2 (Li et al., 2024b), InternVL2 (OpenGVLab, 2024), LLaVA-OneVision (Li et al., 2024a), etc. As detailed in Table 1, CTRL-V demonstrates consistent performance improvements across diverse benchmarks, validating the effectiveness of our alignment strategy.

Temporal Directionality and Causal Consistency. A primary challenge in video understanding is distinguishing temporal directionality. This capability is rigorously evaluated by the **Strict Accuracy** metric on RTime-QA, which credits a model only if it correctly answers questions for *both* the forward and reversed video clips. CTRL-V achieves a Strict Accuracy of **73.48%**, significantly surpassing the previous best performer VideoChat2 (54.26%) and widely used models like LLaVA-OneVision (39.42%). This substantial margin indicates that our RL-based contrastive optimization effectively **mitigates the temporal ambiguity** often prevalent in baselines, enabling the model to form a consistent causal representation. This robustness extends to TemporalQA, where CTRL-V achieves **75.76%** accuracy, further confirming its ability to handle complex queries regarding event evolution. This confirms that CTRL-V does not merely fit the data but fundamentally sharpens the model's **sensitivity to the temporal directionality**.

Generalization to Temporal-Aware Comprehension. Beyond specific tasks, the learned temporal logic translates effectively to general video comprehension and captioning. On TempCompass, a benchmark demanding high sensitivity to event sequences, CTRL-V achieves **67.56%** which outperforms strong baselines like InternVL2 and MiniCPM-V 2.6 (Yao et al., 2024). This suggests that the model generates more temporally coherent descriptions. Furthermore, on MVBench, a comprehensive suite for fine-grained action and object perception, our CTRL-V achieves **68.03%**, exceeding specialized long-video models like LongVU (Shen et al., 2025). These results collectively show that enhancing temporal sensitivity can hardly compromise spatial understanding. In contrast, it fosters a more holistic video representation that benefits tasks ranging from fine-grained recognition to open-ended narration.

5.4. Effectiveness across various backbones

We evaluate the robustness of CTRL-V by applying it to three Video-LLMs: Qwen2.5-VL-7B (Bai et al., 2025b), LLaVA-NeXT-Video-7B (Zhang et al., 2024b), and Qwen3-VL-8B (Bai et al., 2025a). Operating in a strictly label-free manner, our framework delivers substantial improvements across diverse backbones without accessing ground-truth annotations. Notably, LLaVA-NeXT achieves a 10.57% accuracy boost on TemporalQA, while the advanced Qwen3-VL records a 7.12% increase on MVBench. These consistent gains across benchmarks (Table 2) confirm that CTRL-V effectively unlocks latent temporal capabilities and scales seamlessly, elevating performance even in already competitive architectures.

5.5. Abalation Study

Self-Supervision Yields Superior Generalization. As presented in Table 3, a critical comparison lies between CTRL-V and Rejection Sampling Fine-Tuning (RFT). The disparity in supervision is fundamental: RFT acts as an oracle by filtering samples with ground truth labels, while CTRL-V derives signals purely from intrinsic contrast. This difference manifests distinctly in the numerical results. On the source domain (TemporalQA), RFT naturally outperforms CTRL-V (80.30% vs. 75.76%) due to its direct access to gold-standard annotations. However, on all three unseen benchmarks (RTime-QA, TempCompass, MVBench), **CTRL-V consistently surpasses RFT** (e.g., 85.16% vs. 77.13% on RTime-QA). This reversal validates our theoretical premise: while RFT merely fits the training distribution, CTRL-V structurally **disentangles causal dynamics**. By penalizing semantic counterfactuals, it effectively elicits latent reasoning capabilities, resulting in a robust inference mechanism that transfers more effectively to unseen domains.

Table 3. Ablation study of the proposed CTRL-V on four temporal video QA benchmarks. Best results are highlighted in bold.

Model	RTime-QA			TempCompass	
	Acc	Strict	Score	Acc	Score
w/o all	68.49	50.36	3.64	62.13	3.61
RFT	77.13	62.29	3.95	67.17	3.69
SFT	68.61	46.96	3.31	59.93	3.36
Base + CD	69.10	50.36	3.66	66.82	3.69
CTRL-V	**85.16**	**73.48**	**4.29**	**67.56**	**3.72**

Model	MVBench		**TemporalQA**	
	Acc	Score	Acc	Score
w/o all	62.01	3.69	64.87	3.64
RFT	64.52	3.82	**80.30**	**4.27**
SFT	58.32	3.36	67.01	3.56
Base + CD	65.89	3.80	68.57	3.87
CTRL-V	**68.03**	**3.95**	75.76	4.04

Superiority of Internalized Learning over Inference Heuristics. The decisive performance gap between **CTRL-V** and *Base+CD* on *RTime-QA* (85.16% vs. 69.10%) underscores the necessity of embedding temporal logic directly into the model's parameters. While *Base+CD* offers a negligible 0.61% improvement over the baseline through post-hoc decoding adjustments, our **CTRL-V** achieves a substantial 16.06% absolute gain. This disparity proves that superficial probability shifts during inference cannot substitute for fundamental representation learning. By enforcing self-consistency during training, the proposed **CTRL-V** effectively reshapes the latent space, transforming temporal causality from an external heuristic into an intrinsic capability of the model weights.

Temporal Reversal is the Superior Negative Constraint. As illustrated in Figure 3, the strategy of using temporally reversed video (V_{rev}) significantly outperforms random shuffling and noise injection baselines. Unlike random shuffling,

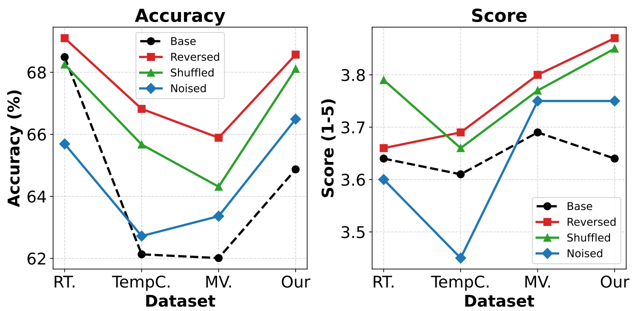

Figure 3. Effect of Different Negative Sampling Strategies.

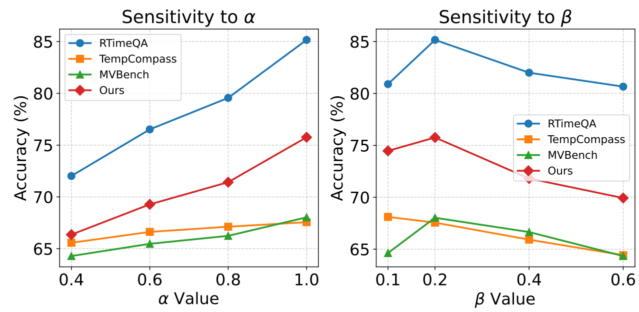

Figure 4. Sensitivity analysis of hyperparameters α and β.

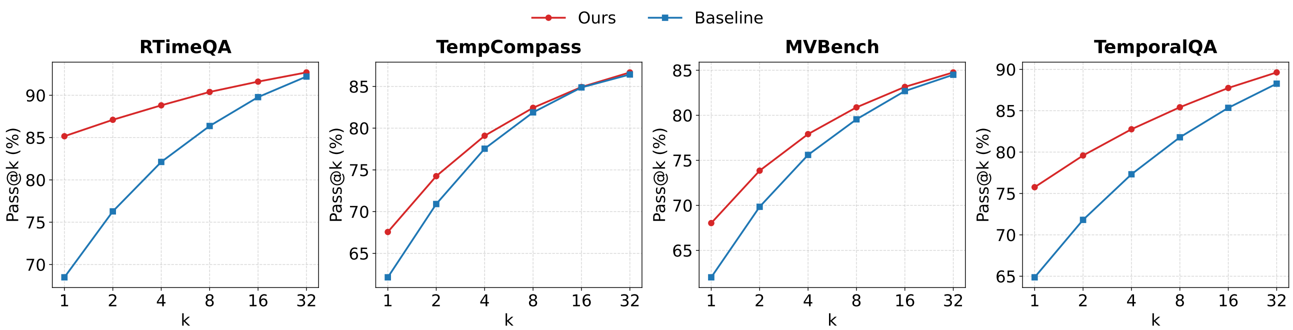

Figure 5. Pass@k evaluation on four benchmarks, including RTime-QA, TempCompass, MVBench, and TemporalQA. Our framework effectively excavates and activates the model's latent temporal reasoning capabilities through self-consistent supervision.

which destroys temporal coherence and leads to semantic instability, the reversed stream maintains high visual plausibility while establishing a **precise semantic contradiction**. This forces the model to rely solely on the temporal directionality for discrimination, thereby ensuring that the learned features are genuinely temporal without human annotations.

Impact of Rectification Strength and Safety Truncation. Figure 4 details the sensitivity of the contrastive hyperparameters. We observe a monotonic performance gain as the rectification strength α increases, confirming the necessity of **strongly rectifying** the inverse distribution to suppress anti-causal hallucinations. Regarding the safety threshold, performance peaks at $\beta = 0.2$. Lower values fail to filter noise, while higher values impose excessive truncation, inadvertently excluding valid tokens that are necessary for correct prediction. This indicates that a moderate "safety valve" is essential for balancing the trade-off between suppressing hallucinations and maintaining lexical diversity.

5.6. Validation of Intrinsic Reasoning Elicitation

The `pass@k` trajectories presented in Figure 5 provide compelling empirical corroboration for our theoretical analysis in Section 3. Specifically, the convergence of performance at higher k values serves as a critical validation of our "latent excavation" hypothesis rather than indicating a saturation of capability. This trend confirms that the baseline model indeed possesses the requisite temporal knowledge, yet this knowledge remains submerged in the long tail of the probability distribution due to inherent ambiguity. By achieving substantial performance margins at low k (e.g., $k = 1, 2$), the proposed CTRL-V demonstrates its efficacy in sharpening the likelihood profile and promoting these correct but initially low-confidence hypotheses to the dominant mode. This aligns perfectly with our label-free training paradigm, proving that our framework functions by disentangling temporal ambiguity to activate the model's intrinsic potential, rather than by injecting external supervision.

6. Conclusion

In this work, we introduced **CTRL-V** to address temporal ambiguity by treating reversed video streams as semantic counterfactuals. By leveraging reinforcement learning on these intrinsic contrasts, the model enhances its causal reasoning capabilities without the need for external supervision. This success points towards a broader potential framework **CTRL-X**. Extensive experiments show the effectiveness of the proposed method. Future research can explore flexible definitions of "X" by devising diverse negative samples or employing alternative training-free guidance strategies. Such expansions would enable the adaptive refinement of model logic across varied downstream tasks.

Impact Statement

We do not anticipate specific negative societal consequences beyond the general risks associated with generative AI.

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A. Detailed Derivation of Theoretical Analysis

In this appendix, we provide the complete mathematical derivation for the theoretical claims made in Section 3. We rigorously detail the construction of the contrastive log-probability, the step-by-step decomposition of the causal decision margin, and the gradient dynamics that drive the self-reinforcing loop.

A.1. Problem Setup and Definitions

Let $\mathcal{V}$ denote the vocabulary space. For a given video V and query q, the model parameterized by θ generates a probability distribution $\pi_\theta(y_t|V, q, y_{<t})$ at step t.

We define the following entities:

- V^+: The original video sequence.
- V^-: The **negative sample** constructed via a specific transformation Φ, designed to represent a temporal flow inconsistent with the ground truth.
- y^*: The ground-truth token corresponding to the actual event in V^+.
- y^-: The **alternative token** representing the semantic content corresponding to V^-, which is semantically distinct from and inconsistent with y^*.

Assumption. We assume a standard model often struggles to distinguish subtle temporal variations solely based on visual cues. Mathematically, it assigns a non-negligible probability to the inconsistent token y^- in the consistent context V^+:

$$\pi_\theta(y^-|V^+) > \epsilon, \tag{13}$$

where ϵ is a threshold indicating ambiguity.

A.2. Derivation of the Rectified Contrastive Log-Probability

As defined in Eq. 5 of the main text, the rectified log-probability $\mathcal{L}_t(w)$ for a token w is constructed to penalize the inconsistent semantics:

$$\mathcal{L}_t(w) = (1+\alpha)\ln \pi_\theta(w|V^+) - \alpha \ln \pi_\theta(w|V^-). \tag{14}$$

(Note: Conditioning on q and $y_{<t}$ is omitted for brevity).

Proposition 1 (Differential Gain). *The log-probability function $\mathcal{L}$ maximizes the gap between the true positive y^* and the hard negative y^-.*

Proof. We examine the asymptotic behavior of the divergence $\Delta\mathcal{L}(y) = \mathcal{L}(y^*) - \mathcal{L}(y^-)$.

1. For the ground truth y^*: The original context V^+ visually supports y^*, while the negative context V^- contradicts it. Thus, as $\pi(y^*|V^+) \to 1$ and $\pi(y^*|V^-) \to 0$:

$$\mathcal{L}(y^*) \approx (1+\alpha)\cdot \ln(1) - \alpha \cdot \ln(\epsilon) \approx -\alpha \ln(\epsilon) \gg 0. \tag{15}$$

2. For the alternative token y^-: The original context V^+ provides weak support, but the negative context V^- strongly supports y^- as the correct description of the transformed input. Thus, as $\pi(y^-|V^-) \to 1$:

$$\mathcal{L}(y^-) \approx (1+\alpha)\ln(\delta) - \alpha \cdot \ln(1) \approx (1+\alpha)\ln(\delta) \ll 0. \tag{16}$$

where δ represents the probability of the alternative semantic in the original pass.

Therefore, the likelihood ratio between selecting y^* versus y^- under the Softmax operation becomes exponentially large:

$$\frac{P(y^*)}{P(y^-)} = \frac{e^{\mathcal{L}(y^*)}}{e^{\mathcal{L}(y^-)}} = e^{\mathcal{L}(y^*)-\mathcal{L}(y^-)} \to \infty. \tag{17}$$

This proves that the contrastive mechanism effectively filters hard negatives by exploiting the semantic contradiction in the negative stream. □

A.3. Detailed Decomposition of the Decision Margin

Here we provide the step-by-step algebraic expansion for the decomposition of the Contrastive Margin $\mathcal{M}(\theta)$, which justifies the dual-objective nature of our framework.

Definition. The margin is defined as the expected log-probability difference between the ground truth and the alternative semantic:

$$\mathcal{M}(\theta) = \mathcal{L}(y^*) - \mathcal{L}(y^-). \tag{18}$$

Step 1: Substitution. Substitute the definition of $\mathcal{L}$ into the margin equation:

$$\mathcal{M}(\theta) = \left[(1+\alpha)\ln \pi(y^*|V^+) - \alpha \ln \pi(y^*|V^-)\right] - \left[(1+\alpha)\ln \pi(y^-|V^+) - \alpha \ln \pi(y^-|V^-)\right]. \tag{19}$$

Step 2: Regrouping by Coefficients. We group terms associated with the positive weight $(1+\alpha)$ and the negative penalty α:

$$\mathcal{M}(\theta) = \underbrace{(1+\alpha)\ln \pi(y^*|V^+) - (1+\alpha)\ln \pi(y^-|V^+)}_{\text{Term A: Positive Context Components}} + \underbrace{\left(-\alpha \ln \pi(y^*|V^-)\right) - \left(-\alpha \ln \pi(y^-|V^-)\right)}_{\text{Term B: Negative Context Components}}. \tag{20}$$

Step 3: Handling the Sign Flip. In Term B, observe that subtracting a negative is equivalent to adding. We factor out α:

$$\text{Term B} = -\alpha \ln \pi(y^*|V^-) + \alpha \ln \pi(y^-|V^-) = \alpha\left[\ln \pi(y^-|V^-) - \ln \pi(y^*|V^-)\right]. \tag{21}$$

Crucial Observation: The subtraction order flips in the negative context term (y^- comes first), because y^- was the token being penalized (subtracted) in the original definition.

Step 4: Logarithmic Transformation. Applying the identity $\ln a - \ln b = \ln \frac{a}{b}$ to both Term A and Term B:

$$\mathcal{M}(\theta) = (1+\alpha)\ln\left(\frac{\pi(y^*|V^+)}{\pi(y^-|V^+)}\right) + \alpha \ln\left(\frac{\pi(y^-|V^-)}{\pi(y^*|V^-)}\right) = \mathcal{M}_{\text{pos}}(\theta) + \mathcal{M}_{\text{neg}}(\theta). \tag{22}$$

This concludes the rigorous derivation of Eq. 7 presented in the main text.

A.4. Analysis of Optimization Dynamics

Finally, we analyze why the GRPO update leads to a monotonic increase in $\mathcal{M}(\theta)$, creating the self-reinforcing loop.

The reinforcement learning objective $\mathcal{J}$ encourages the policy π_θ to align with the distribution derived from $\mathcal{L}$. Since $\mathcal{L}(y^*) \gg \mathcal{L}(y^-)$, the optimization explicitly increases the likelihood of y^* in the positive context.

We analyze the partial derivative of the margin with respect to the training iteration k:

$$\frac{\partial \mathcal{M}}{\partial k} = (1+\alpha)\frac{\partial \mathcal{M}_{\text{pos}}}{\partial k} + \alpha \frac{\partial \mathcal{M}_{\text{neg}}}{\partial k}. \tag{23}$$

1. **Positive Discriminability ($\mathcal{M}_{\text{pos}}$):** The GRPO update explicitly maximizes the probability of the correct token y^* in V^+. Thus, the log-likelihood ratio $\ln \frac{\pi(y^*|V^+)}{\pi(y^-|V^+)}$ increases.

$$\frac{\partial \mathcal{M}_{\text{pos}}}{\partial k} > 0. \tag{24}$$

2. **Negative Recognition ($\mathcal{M}_{\text{neg}}$):** As the model learns better semantic alignment from the primary task, its generalized understanding improves. A more capable encoder correctly recognizes the negative dynamics in V^-, assigning higher probability to the alternative token y^-.

$$\frac{\partial \pi(y^-|V^-)}{\partial k} > 0 \implies \frac{\partial \mathcal{M}_{\text{neg}}}{\partial k} > 0. \tag{25}$$

Combining these, we have $\frac{\partial \mathcal{M}(\theta)}{\partial k} > 0$. This confirms that the optimization loop is self-reinforcing: improvements in policy capability directly translate to a larger discriminative margin, which in turn generates higher-quality supervision for the next iteration.

A.5. Probabilistic Interpretation of Contrastive Logits

Let $\pi_\theta(y \mid V)$ denote the probability distribution of the base model. The contrastive decoding module modifies the logits $\mathcal{L}_\theta = \ln \pi_\theta$ to generate a pseudo-target. The contrastive logit is defined as $\tilde{\mathcal{L}}(y) = (1+\alpha)\mathcal{L}_\theta(y \mid V^+) - \alpha\mathcal{L}_\theta(y \mid V^-)$. To understand the distributional implication, we transform this linear rectification back into the probability space. By exponentiating both sides, we derive the implicit **Excavated Teacher Distribution** $\pi_{\text{excavated}}$:

$$\begin{aligned} \pi_{\text{excavated}}(y) &\propto \exp\left(\tilde{\mathcal{L}}(y)\right) \\ &= \exp\left(\ln \pi_\theta(y \mid V^+) + \alpha\left(\ln \pi_\theta(y \mid V^+) - \ln \pi_\theta(y \mid V^-)\right)\right) \\ &= \pi_\theta(y \mid V^+) \cdot \left(\frac{\pi_\theta(y \mid V^+)}{\pi_\theta(y \mid V^-)}\right)^\alpha \end{aligned} \tag{26}$$

A.6. The Self-Excavation Identity via Importance Sampling

The GRPO objective encourages the policy to approximate this target. Theoretically, the gradient of minimizing the divergence between the current policy and the target, viewed through the lens of Importance Sampling, is scaled by the density ratio $w(\tau) = \pi_{\text{excavated}}(\tau)/\pi_\theta(\tau \mid V^+)$. Substituting Eq. 26 into this ratio reveals a critical mathematical cancellation:

$$w(\tau) = \frac{\pi_\theta(\tau \mid V^+) \cdot \left(\frac{\pi_\theta(\tau \mid V^+)}{\pi_\theta(\tau \mid V^-)}\right)^\alpha}{\pi_\theta(\tau \mid V^+)} \equiv \underbrace{\left(\frac{\pi_\theta(\tau \mid V^+)}{\pi_\theta(\tau \mid V^-)}\right)^\alpha}_{\Psi(\tau;\theta)} \tag{27}$$

This proves that the effective optimization signal $\Psi(\tau;\theta)$ is independent of the external decoding mechanism's form and is derived solely from the model's intrinsic parameters. Incorporating the trust region constraint, we arrive at the final stabilized gradient formulation presented in the main text:

$$\nabla_\theta \mathcal{J} \cong \mathbb{E}_{\tau \sim \pi_\theta}\left[\mathbb{I}_{\text{clip}}(\rho_\tau) \cdot \Psi(\tau;\theta) \cdot \nabla_\theta \ln \pi_\theta(\tau \mid V^+)\right] \tag{28}$$

B. Our Dataset

As shown in Figure 6, current video instruction-tuning datasets often suffer from a lack of directional specificity in their annotation. By predominantly relying on generic, open-ended prompts (e.g., "Describe the activity in the video"), these standard corpora fail to provide the explicit supervision signals necessary for models to learn temporal reasoning pathways. Training on such broad formulations encourages models to merely aggregate salient visual elements or recite learned semantic associations, without genuinely parsing the chronological structure of events. Consequently, models trained on these datasets often recognize content existence but struggle to internalize the underlying temporal dynamics.

To address this deficiency, we curate **TemporalQA**, a comprehensive dataset specifically constructed to **instill** and evaluate directed temporal reasoning. Organized into training, validation, and test splits, TemporalQA serves as the primary instructional corpus for our model. While retaining an open-ended generation format to preserve natural language versatility, the dataset employs **chronologically targeted prompts** as the core supervisory signal. These prompts require the model to explicitly articulate the topological relationships between events during training, focusing on sequential logic and causal progression—asking not just "what happened," but specifically "what action did the girl perform *subsequently*?" or "what event immediately *preceded* the outcome X?".

By integrating these directive, structure-aware inquiries into the training phase, TemporalQA compels the model to move beyond isolated observation and engage in active temporal reconstruction. The training split provides the necessary inductive bias for the model to resolve the arrow of time, while the test split offers a rigorous assessment of this acquired capability. This design transforms the dataset from a passive collection of descriptions into an active **instructional mechanism**, enabling the model to establish a robust internal representation of causality through targeted supervision.

C. Experiment

C.1. Benchmarks

RTime-QA is a benchmark of 822 human-annotated video-question pairs designed to evaluate atomic temporal event understanding, specifically the ability to distinguish forward versus reverse event progressions. Each test item is built upon a

Figure 6. Illustrative comparison of question design in existing benchmarks vs. TemporalQA.

video pair depicting the same physical scene and objects but playing in opposite temporal directions, such that static frames are nearly identical and only the temporal dynamics differ. In our work, we convert the original multiple-choice format into open-ended QA, prompting models to generate a free-form description of the event in the given video rather than selecting between pre-defined options, thereby requiring robust reasoning over fine-grained temporal directionality.

TempCompass is a benchmark designed to assess fine-grained temporal understanding in Video LLMs across five aspects (e.g., action, speed, direction, event order, and attribute change) and four task formats. Among these, we specifically use the **Caption Generation** task because it is the most challenging. This task requires models to first select the correct temporal information from multiple distractors and then generate a concise and faithful caption, thereby serving as a stringent test of both temporal understanding and generation fidelity.

MVBench is a comprehensive benchmark designed for evaluating the temporal understanding capabilities of multimodal large language models on video inputs. It comprises 20 challenging tasks that cannot be solved using a single frame and span a wide spectrum from perception to high level cognition including action sequence reasoning action prediction object interaction moving direction estimation scene transition recognition and counterfactual inference.This design ensures temporal sensitivity diverse video scenarios and evaluation fairness without reliance on labor intensive manual labeling.

C.2. Implementation and Evaluation Protocol

For video input processing, we uniformly sample 8 frames from each sequence. The model is fine-tuned on the TemporalQA dataset for 2 epochs. Specifically, we set the penalty strength to $\alpha = 1.0$ and the plausibility constraint to $\beta = 0.2$ during fine-tuning. When transferring our framework to other backbone models, we empirically adjust these training configurations to $\alpha = 1.0$ and $\beta = 0.1$ to accommodate varying distribution sensitivities.

We evaluate performance across four benchmarks: RTime-QA, TempCompass, MVBench, and TemporalQA. To ensure consistent and objective assessment, we employ GPT-3.5-turbo to score model responses following a standardized rubric. Specifically for RTime-QA, we report the *strict Accuracy* metric, which counts a prediction as correct only if the model successfully answers both questions in a paired sample (i.e., corresponding to the original and the reversed video). This rigorous criterion emphasizes robust temporal understanding and penalizes inconsistency in frame ordering sensitivity.

C.3. Training Dynamics and Stability.

Optimization Efficiency and Architectural Universality. Figure 8 visualizes the evolution of training metrics across three distinct backbone models. As illustrated in the Center column, the Reward curves exhibit a remarkably consistent

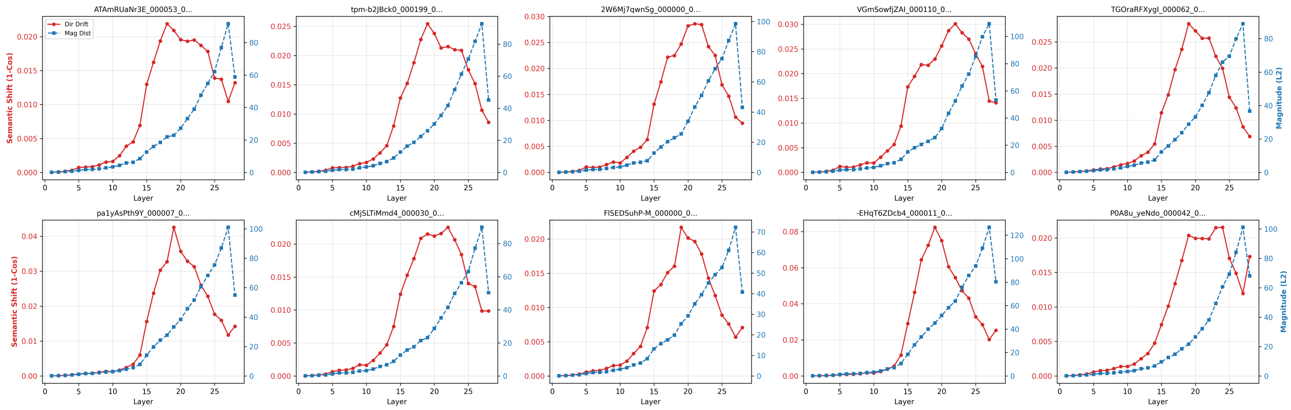

Figure 7. Visualization of internal representation shifts. Comparing the base and fine-tuned models reveals minimal drift in shallow layers, contrasted with significant Semantic Divergence (Red) and Magnitude Shifts (Blue) in deep layers (15–28). This indicates that temporal adaptation occurs primarily within the deep semantic reasoning stages while preserving low-level visual perception.

trajectory characterized by a rapid ascent during the initial training phase, followed by stabilization into a high-value plateau. This steep learning curve suggests that the proposed contrastive signal provides a low-variance, high-fidelity gradient direction, enabling the models to swiftly grasp the underlying temporal logic even in the complete absence of ground-truth supervision. Notably, this convergence pattern is agnostic to the specific architectural nuances of the backbone, indicating that CTRL-V does not rely on serendipitous parameter initializations but rather captures a fundamental mechanism of temporal alignment. The subsequent plateau signifies that the models have effectively saturated the self-supervised objective, maximizing the discriminative margin between the forward causal events and their reversed counterfactuals.

Stability-Plasticity Balance and Convergence Reliability. Simultaneously, the evolution of KL divergence and Reward Standard Deviation (Left and Right columns) demonstrates the stability of our optimization process. The KL divergence increases moderately, reflecting a necessary deviation from the base policy to accommodate new temporal reasoning capabilities (plasticity). Crucially, this metric remains bounded within a safe range, ensuring that the optimization operates within a "trust region" that prevents the catastrophic forgetting of general linguistic competencies (stability). Complementing this, the Reward Standard Deviation exhibits a steady decline or stabilization at low levels. This narrowing of dispersion indicates a transition from stochastic exploration to deterministic execution: as training progresses, the policy becomes increasingly confident in its temporal judgments, reducing the variability of its responses. Together, these dynamics confirm that CTRL-V induces a robust and reliable convergence, successfully instilling temporal precision without compromising the structural integrity of the pre-trained models.

C.4. Effectiveness across backbones

In this subsection, we provide a granular analysis of the performance gains reported in Table 2. By applying CTRL-V across three distinct Video-LLM architectures—Qwen2.5-VL-7B, LLaVA-NeXT-Video-7B, and Qwen3-VL-8B—we observe consistent improvements that highlight the model-agnostic nature of our framework.

The most significant impact of CTRL-V is observed in the **RTime-QA Strict Accuracy** metric, which is particularly unforgiving as it requires the model to correctly identify causal directionality in both forward and reversed video pairs. As shown in Table 2, all three backbones exhibit massive gains, ranging from $+11.19\%$ (LLaVA-NeXT) to a remarkable $+20.20\%$ for Qwen3-VL. The disparity between the modest gains in standard Accuracy and the surge in Strict Accuracy suggests that while base models may correctly answer individual queries by chance, they often lack the intrinsic robustness to distinguish causal directionality consistently. The substantial jump in Qwen3-VL indicates that even advanced models possess a "latent" understanding of time that remains dormant under standard training; our method successfully activates this capability, forcing the model to explicitly reason about the temporal directionality rather than relying on superficial correlations.

Furthermore, the results on **LLaVA-NeXT-Video-7B** highlight the efficacy of our method in architectures that possess strong spatial perception but weaker temporal alignment. LLaVA-NeXT achieves the highest relative gain on the **TemporalQA**

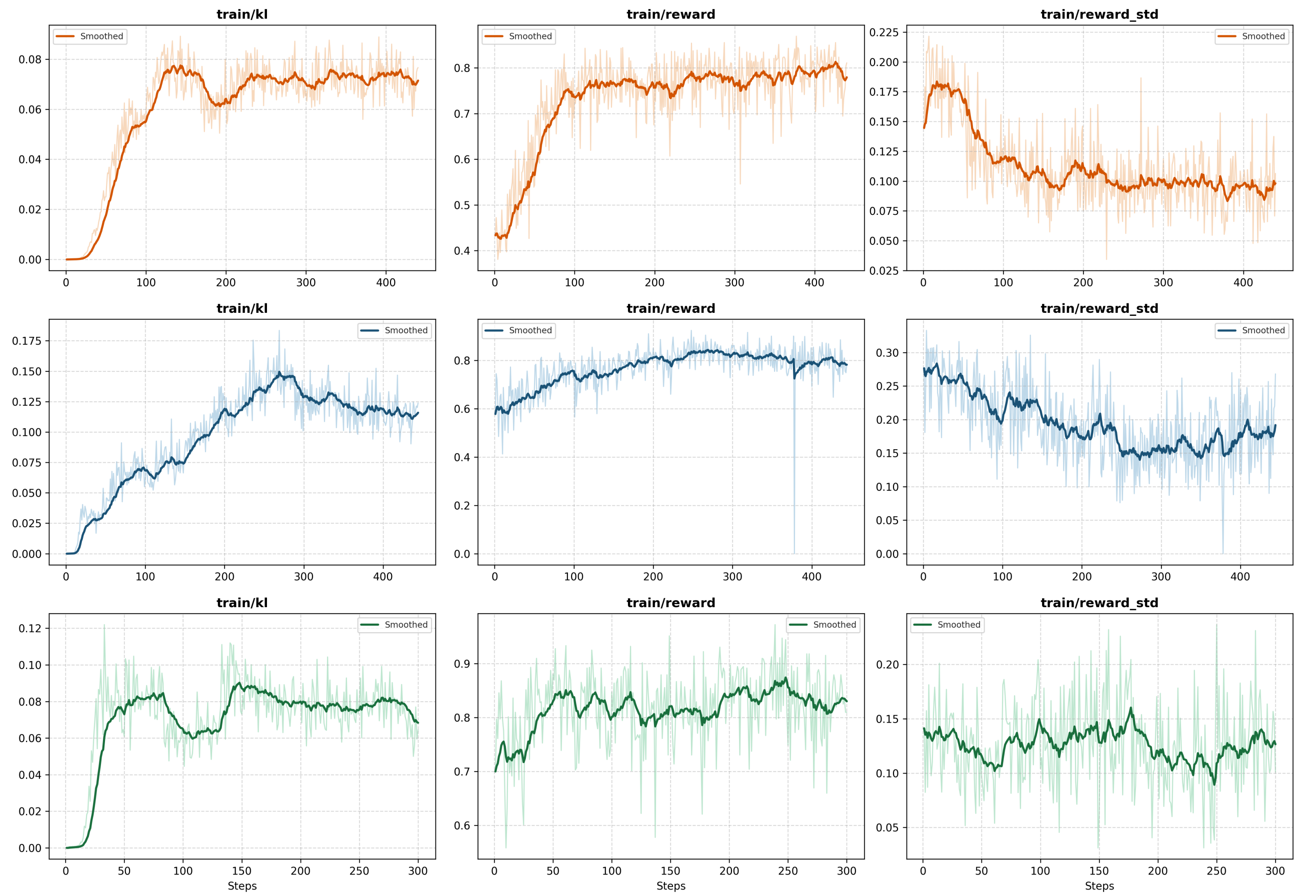

Figure 8. Training dynamics of different Video-LLMs during the RL fine-tuning stage. The rows correspond to LLaVA-NeXT-Video-7B (Top, Orange), and Qwen3-VL-8B (Middle, Blue), Qwen2.5-VL-7B (Bottom, Green). The columns display the evolution of KL divergence (Left), Reward (Center), and Reward Standard Deviation (Right) over training steps. Solid lines represent smoothed values (exponential moving average), while shaded areas indicate the raw fluctuation of the metrics.

dataset, improving from 58.26% to 68.83% ($+10.57\%$). Given LLaVA-NeXT's robust visual encoding, this improvement suggests that its primary limitation was not in perceiving the events, but in sequencing them linguistically. CTRL-V acts as a specialized alignment mechanism, bridging the gap between high-quality visual features and temporal reasoning. This proves that high-quality temporal supervision can be excavated directly from the model's own visual encoder without requiring external architectural changes.

Finally, the performance on **Qwen3-VL-8B** demonstrates that CTRL-V remains effective even for state-of-the-art architectures that already perform well on general benchmarks. Despite Qwen3-VL's strong baseline, CTRL-V delivers a $+7.12\%$ boost on **MVBench** (52.53% to 59.65%), a comprehensive benchmark covering diverse video tasks. The fact that CTRL-V enhances performance on such a general benchmark implies that temporal reasoning is a foundational capability that supports broader cognition. By sharpening the model's sensitivity to causal dynamics, we inadvertently improve its performance on related tasks such as action recognition and event anticipation, confirming that CTRL-V enhances the fundamental cognitive fidelity of the video encoder.

C.5. Mechanistic Analysis

To elucidate the source of performance gains, we conduct a mechanistic analysis of internal representation shifts by visualizing the layer-wise hidden state drift between the base and fine-tuned models (Figure 7). Given that the visual encoder remains frozen, this analysis effectively isolates how the LLM adapts its processing hierarchy to accommodate temporal reasoning.

In the shallow layers (0–10), the hidden states exhibit negligible semantic drift. This stability is pivotal: it suggests that the model preserves the foundational alignment between the fixed visual projections and the LLM's embedding space, safeguarding low-level perception against overfitting. Conversely, a marked surge in divergence appears in the intermediate layers (15–22), identifying this region as the **cognitive core** for temporal adaptation. Here, the model actively restructures its processing of frame-level features to resolve long-term dependencies and verify causal logic. Finally, the pronounced increase in L2 distance within the deep layers (23–28) implies a fundamental **realignment of the output distribution**. This indicates that the model has diverged from the base model's ambiguity, enforcing decisive, structure-aware corrections to the generation of temporal narratives.

D. Case Study

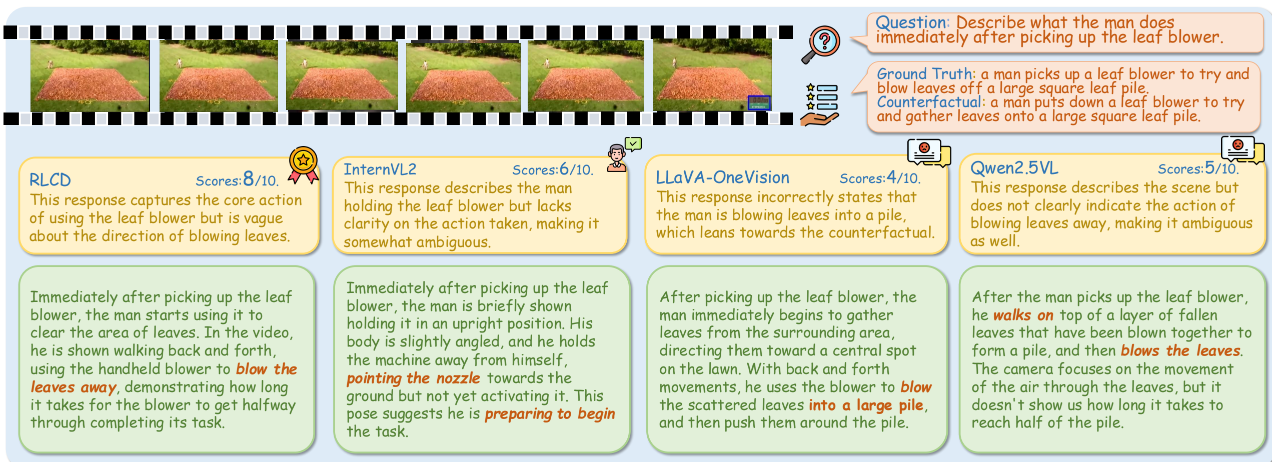

Figure 9. A case study on the TemporalQA dataset, comparing the outputs of four models on a representative time-sensitive QA example. This example illustrates differences in temporal expression recognition and reasoning abilities across models.

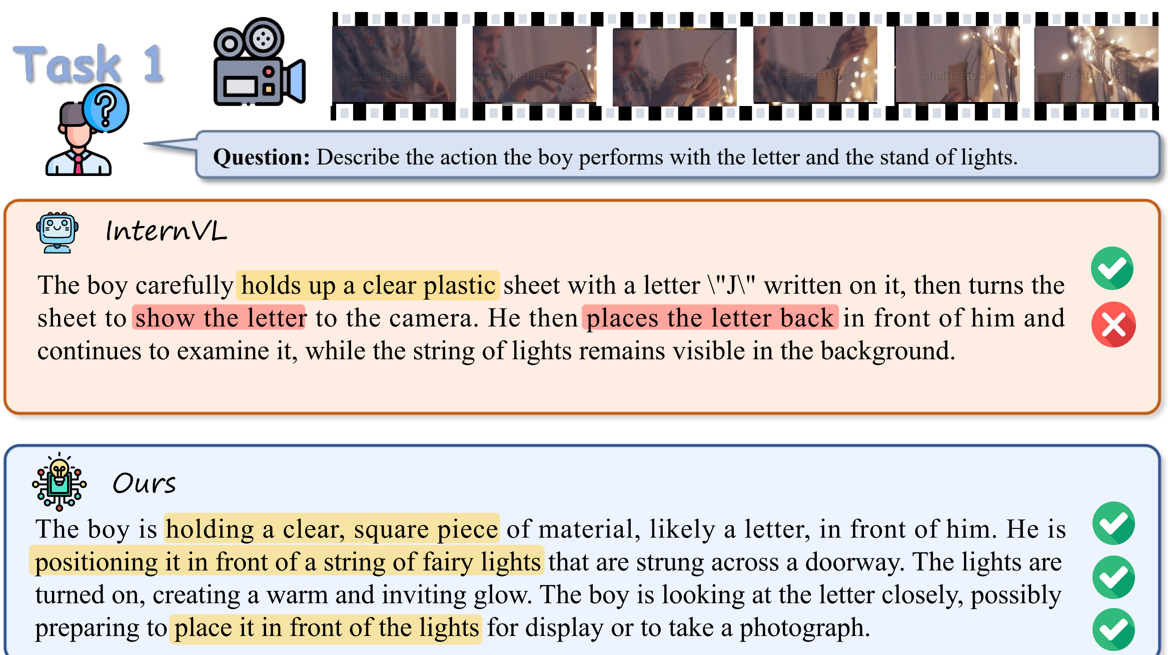

Figure 10. A case study on the RTime-QA dataset, comparing the outputs of two models on a representative time-sensitive QA example. This example illustrates differences in temporal expression recognition and reasoning abilities across models.

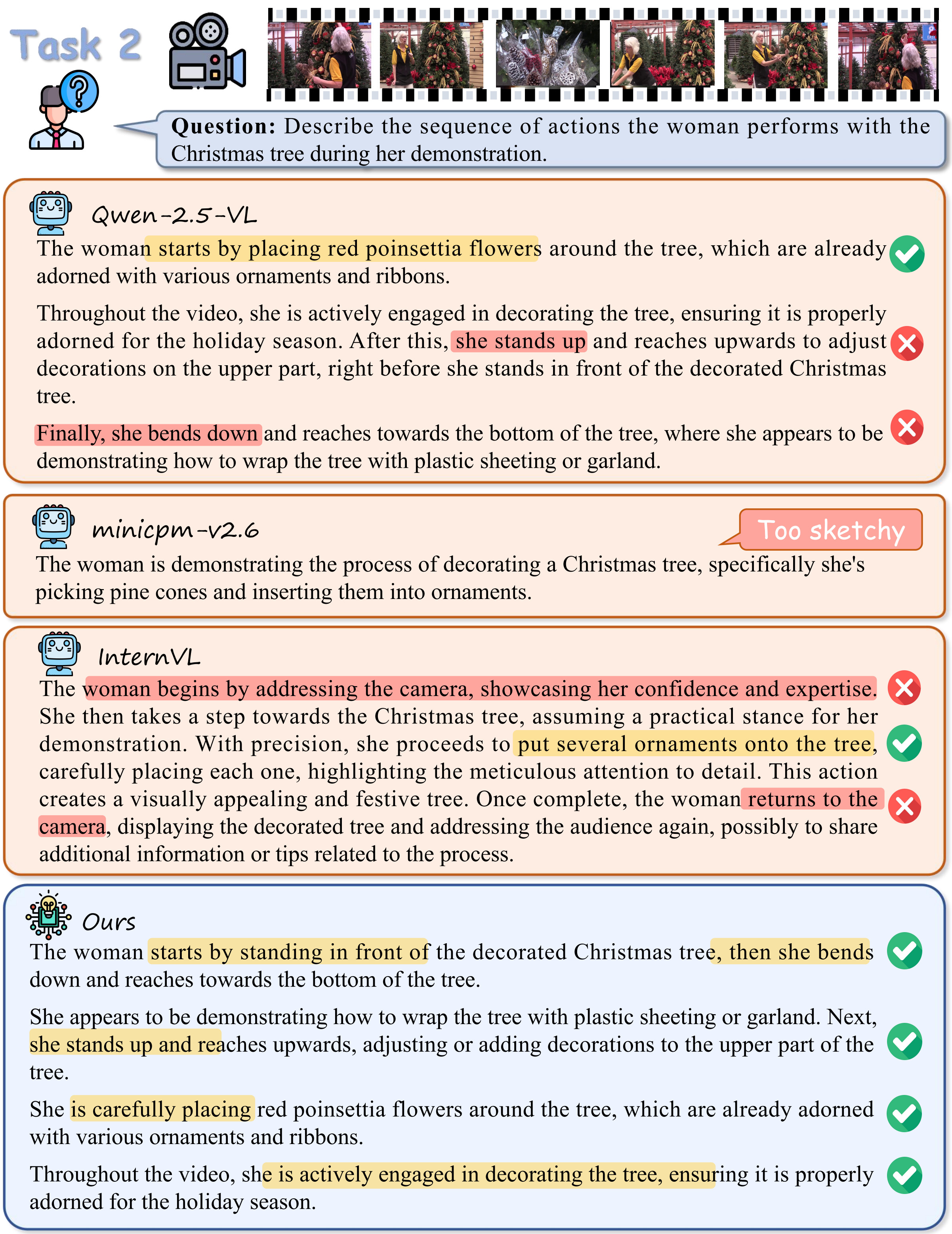

Figure 11. A case study on the TemporalQA dataset, comparing the outputs of four models on a representative time-sensitive QA example. This example illustrates differences in temporal expression recognition and reasoning abilities across models.

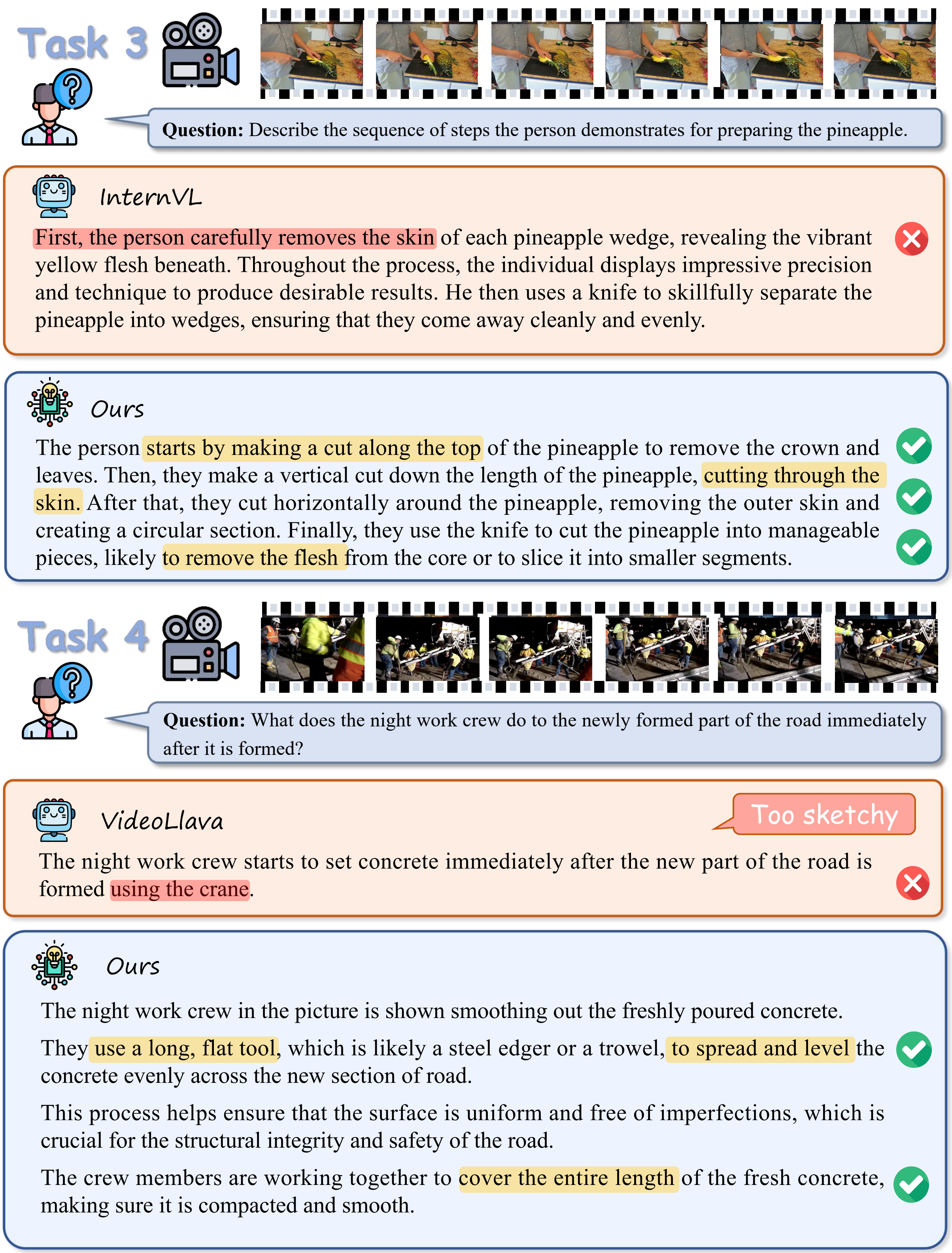

Figure 12. A case study on the TemporalQA dataset, comparing the outputs of three models on a representative time-sensitive QA example. This example illustrates differences in temporal expression recognition and reasoning abilities across models.

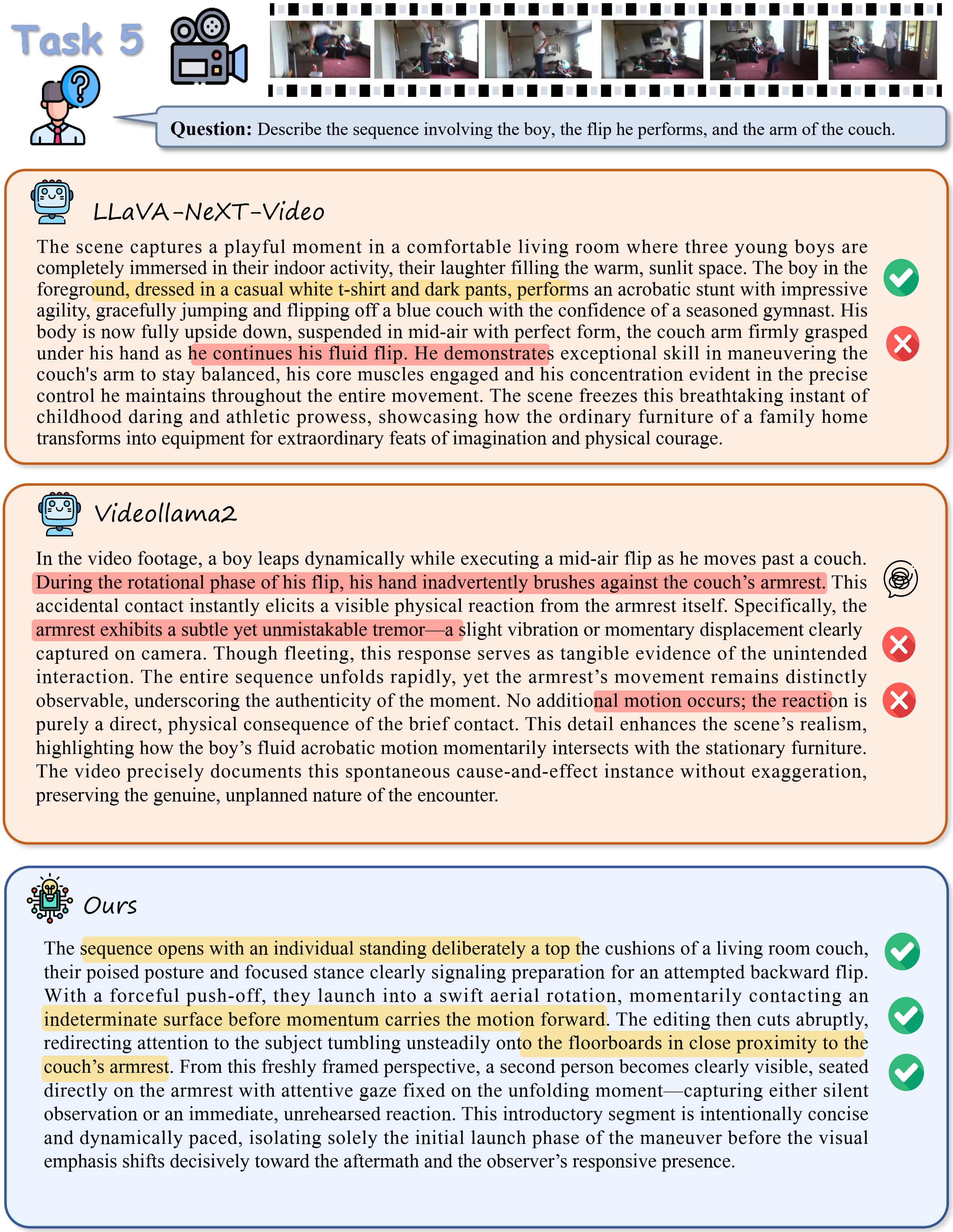

Figure 13. A case study on the TemporalQA dataset, comparing the outputs of three models on a representative time-sensitive QA example. This example illustrates differences in temporal expression recognition and reasoning abilities across models.

Task 6

Question: Describe the sequence of actions the woman demonstrates when making the rose arrangement.

Videollama2

The woman demonstrates several actions when making the rose arrangement, which I will describe in sequence. First, she cuts some of the stems of the roses to the desired length using a pair of scissors. Then, she trims the leaves from the stems, likely to improve the overall appearance and longevity of the bouquet.

Next, she begins to wrap the stems together, possibly with floral tape or ribbon, to hold the arrangement together securely.

Finally, after securing the stems, she wraps greenery around the entire arrangement to add volume, texture, and a more natural look to the finished rose bouquet.

Videollava-7B

Begin by meticulously arranging the individual flower petals sourced directly from the bouquet into a neat, intentionally composed pattern. Focus on visual harmony through thoughtful consideration of symmetry, color flow, petal orientation, and spatial balance to ensure the final layout is aesthetically pleasing. INSTANT STEP: Carefully insert slender floral wires into each petal to provide subtle structural support. Then, secure the wires firmly in place using bobby pins or other preferred fasteners as desired, holding the petals precisely within the arranged design without displacement. Apply fasteners discreetly to preserve the arrangement's delicate appearance while ensuring stability. This technique maintains the integrity of the composition, allowing the petals to remain securely positioned and visually cohesive throughout display or handling.

Ours

The woman in the image is shown actively engaged in creating a bouquet, specifically by trimming the stems. Here a breakdown of her actions:

1. Holding Tools: She has a pair of shears in one hand and appears to be using them

2. Preparing Stems: She carefully trims the green stems of the red roses to prepare them for placement in the vase or other bouquet components.

3. Working with the Roses: The roses themselves, which are part of the bouquet, are already cut and displayed prominently on the table, suggesting that they have been prepared earlier.

Figure 14. A case study on the TemporalQA dataset, comparing the outputs of three models on a representative time-sensitive QA example. This example illustrates differences in temporal expression recognition and reasoning abilities across models.

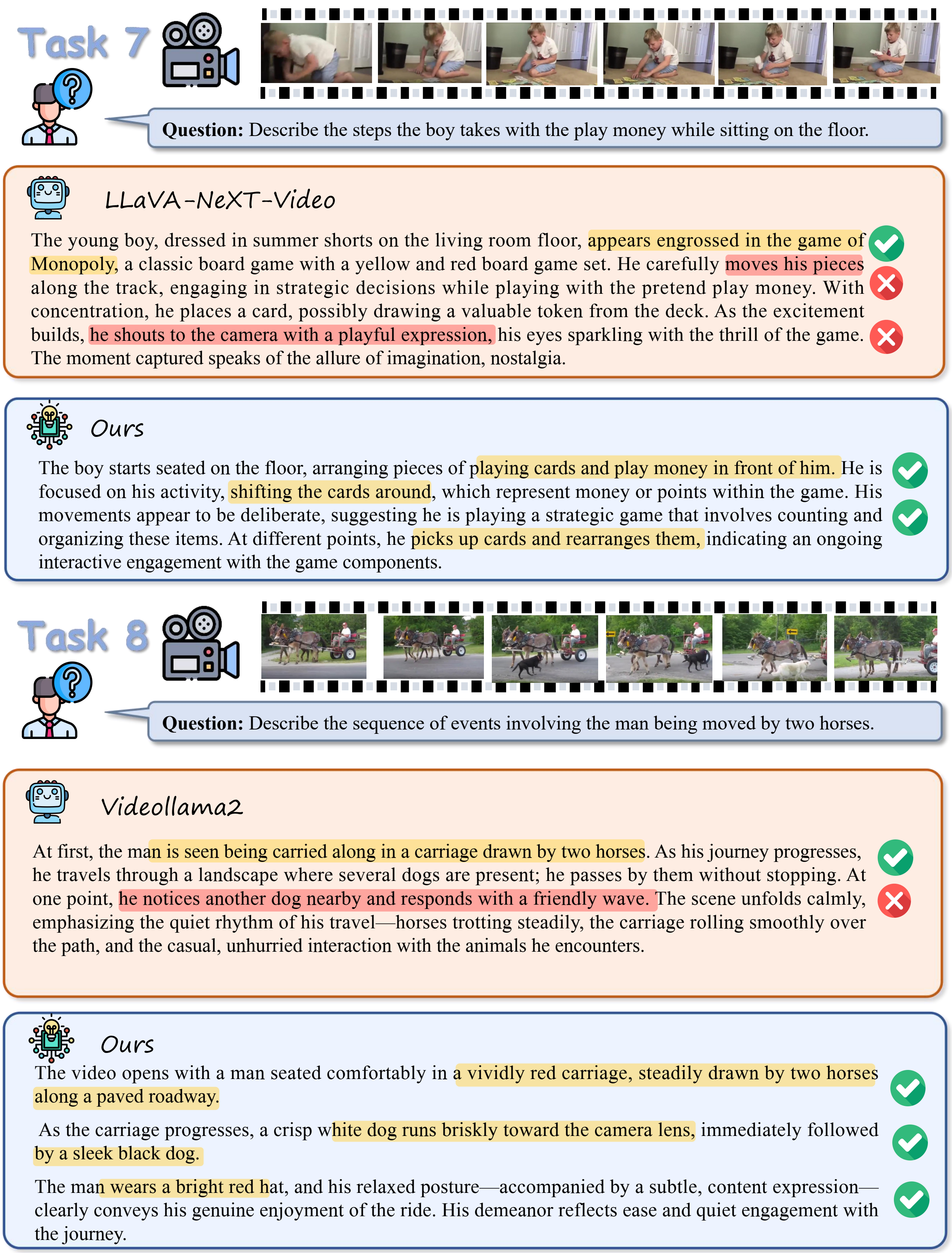

Figure 15. A case study on the TemporalQA dataset, comparing the outputs of three models on a representative time-sensitive QA example. This example illustrates differences in temporal expression recognition and reasoning abilities across models.

Question: Describe the sequence of actions they take to prepare and plant the tree into the ground.

LLaVA-NeXT-Video

The man and woman in the image are preparing to plant a tree in a park. Here's a sequence of actions they take:

1. The man is wearing a red shirt and blue jeans, and he is kneeling down next to the tree. He is holding a shovel and a bag of soil. ✔

2. The woman is wearing a white shirt and black pants, and she is standing next to the man. She is holding a bag of mulch. ✘

3. The man uses the shovel to dig a hole in the ground, creating a circular shape ✘

Videollava-7B

The planting process begins as the man and woman carefully remove the young tree from its pot and set it gently on the prepared ground. ✘

Next, the woman firmly holds the tree upright to keep it stable while the man uses a shovel to dig several small holes in the surrounding soil with deliberate motions.

Finally, the man carefully lowers the tree into the hole, making subtle adjustments to ensure it stands straight and centered, then meticulously backfills the area around the base with the excavated soil. Throughout this coordinated sequence, their focused movements and attentive handling reflect a methodical approach. Each step is carried out with precision to secure the tree firmly in its new position, emphasizing care and teamwork without altering the essential actions of the planting procedure. ✘ ✘

Ours

The video shows a sequence where first, the man and his companions seem to dig a hole. The next action involves the man wearing a red shirt putting on gloves, which suggests they are getting ready for more physical labor associated with the tree planting. ✔

The woman in the blue shirt appears more stationary, possibly assisting or supervising. Finally, the main goal is achieved with them planting the tree firmly into the prepared ground. This process likely involved bringing dirt up into a mound around the base of the tree, a common practice to support newly planted trees. ✔ ✔

Figure 16. A case study on the TemporalQA dataset, comparing the outputs of three models on a representative time-sensitive QA example. This example illustrates differences in temporal expression recognition and reasoning abilities across models.

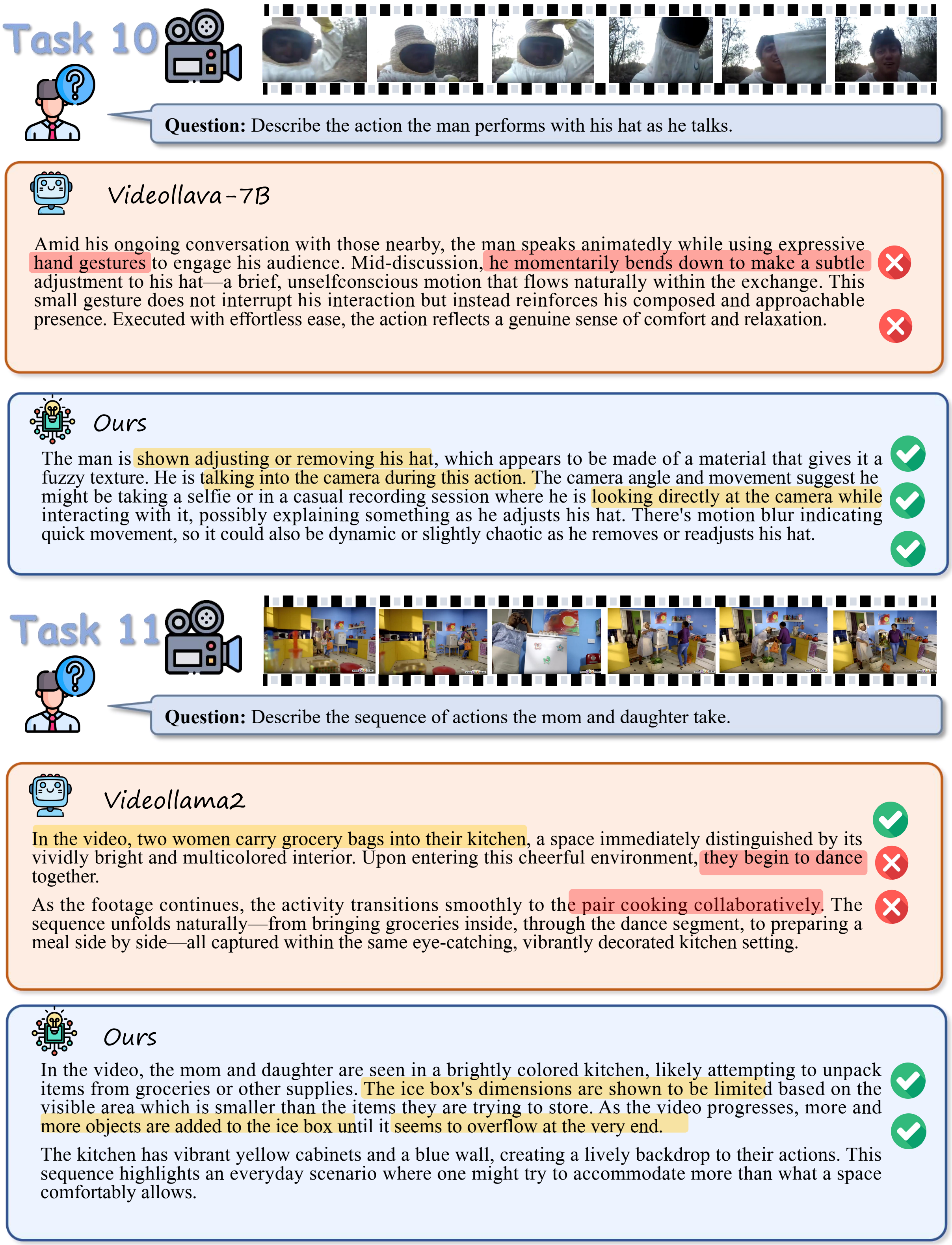

Figure 17. A case study on the TemporalQA dataset, comparing the outputs of three models on a representative time-sensitive QA example. This example illustrates differences in temporal expression recognition and reasoning abilities across models.

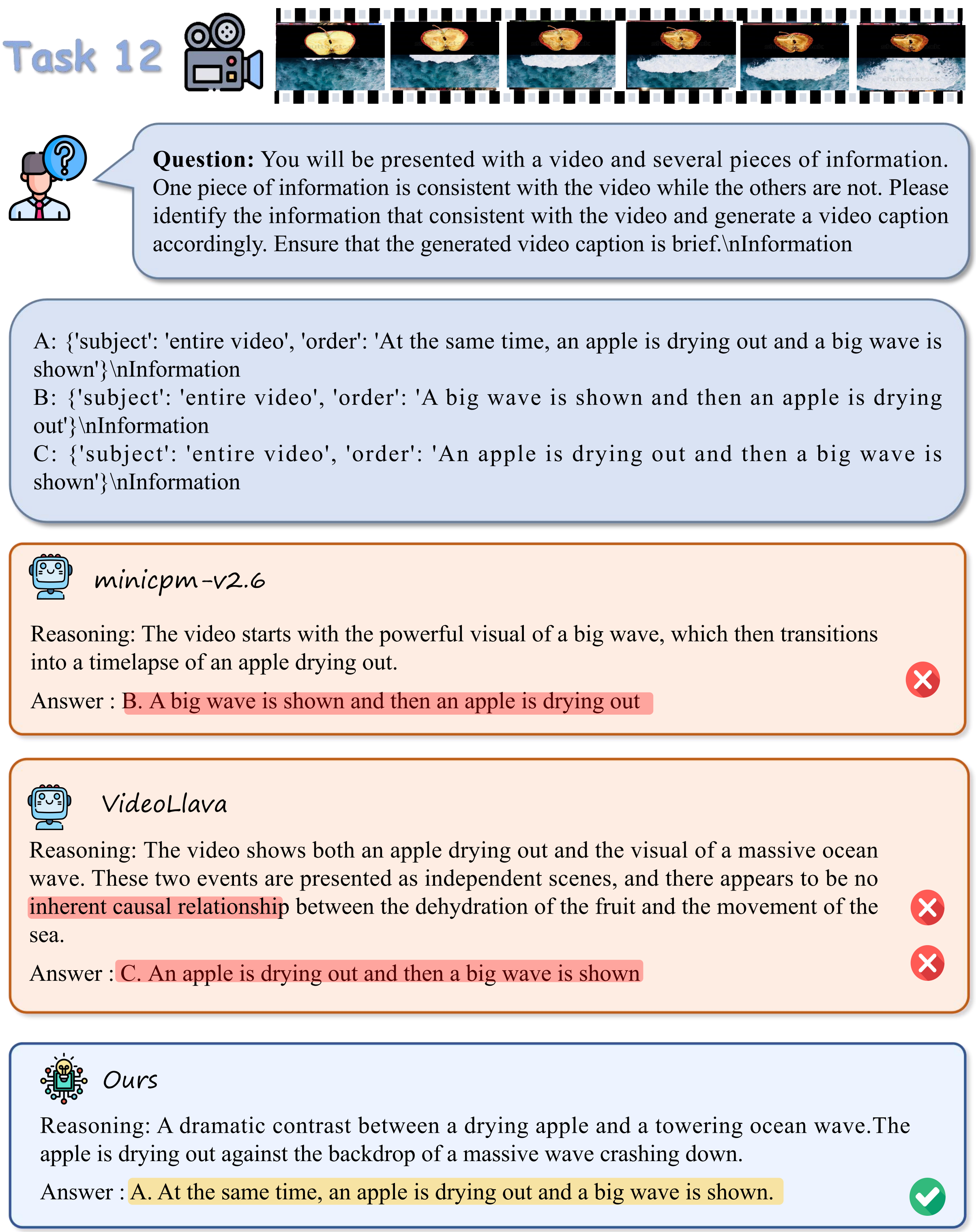

Figure 18. A case study on the TempCompass dataset, comparing the outputs of three models on a representative time-sensitive QA example. This example illustrates differences in temporal expression recognition and reasoning abilities across models.

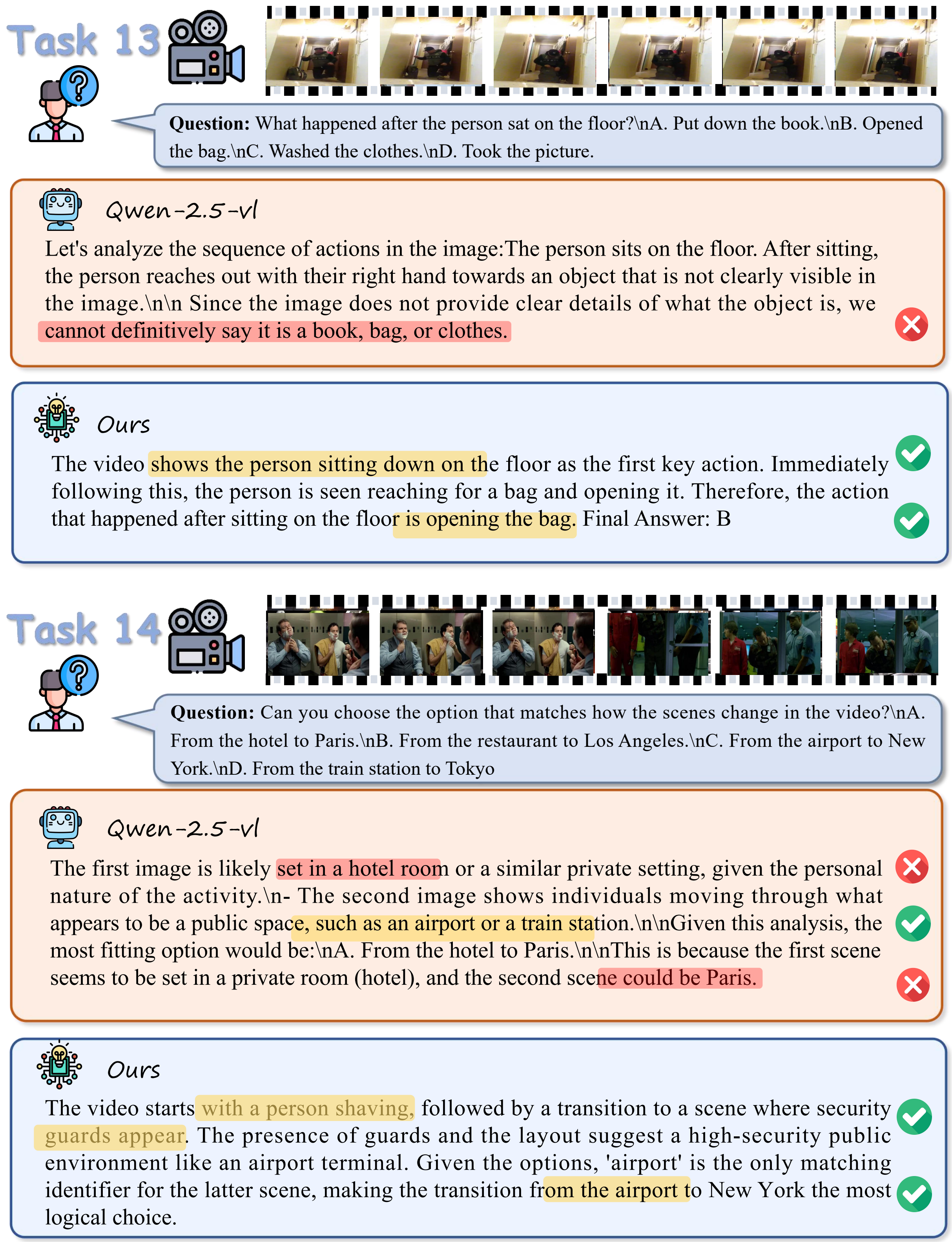

Figure 19. A case study on the MVBench dataset, comparing the outputs of two models on a representative time-sensitive QA example. This example illustrates differences in temporal expression recognition and reasoning abilities across models.